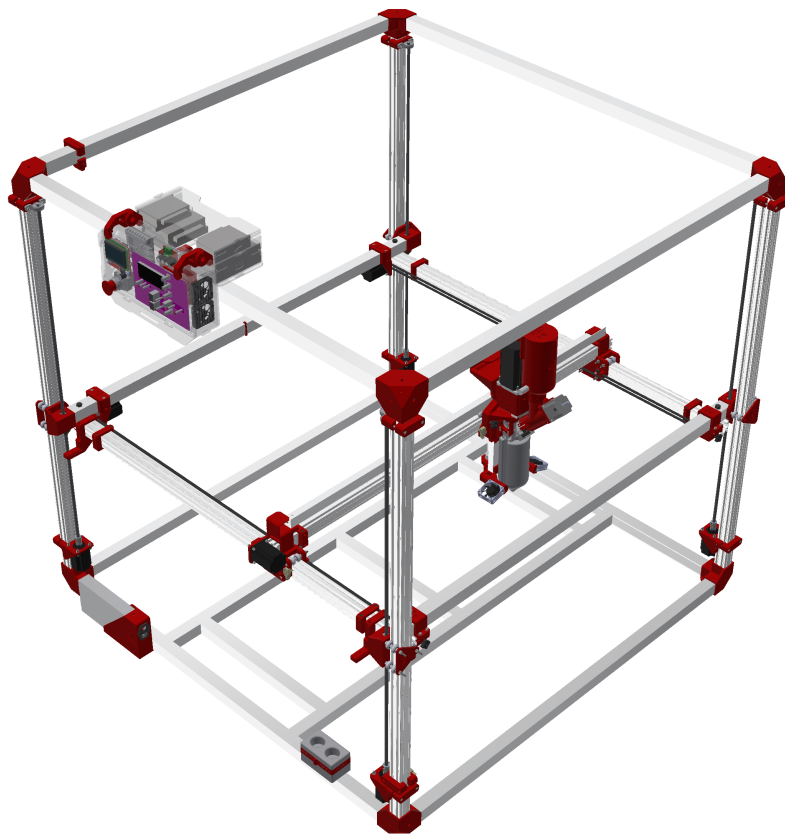


ginger.*one* v1.3 *beta*
the affordable large-scale pellet 3d printer



TRANSLATED MANUAL

PREFACE

The present manual “ginger.one v1.3 beta – Translated Manual” has been written in accordance to the normative UNI EN ISO 20607, " Safety of machinery - Instruction handbook - General drafting principles."

The manual has been written and reviewed at the best of the manufacturer ability

The Manufacturer is not responsible for its usage and reserves the right to make changes to the product and to the information herein provided, without the need to notify the user.



REQUIREMENT: No part of the present document may be copied, duplicated, reproduced, stored with any data archiving system, translated in any language or transmitted with any communication system, without the consent of the Manufacturer.

ABOUT THE MANUFACTURER

Ginger Additive s.r.l.s.

Via Vittorio Veneto 44

20093 San Maurizio al Lambro (MI) – Italia

www.gingeradditive.com

Ginger Additive develops, produces and sells machinery for large scale pellet 3d printing.

The mission statement of the company is to develop technologies in digital fabrication so as to make them accessible in terms of economic opportunity and capabilities for a daily concrete use.

DOCUMENT VERSION

The following table shows the current version of the document and all other available versions:

ES	DA	DE	EL	EN	FR	IT	NL	PT	FL	SV	CS	ET	LV	LT	HU	MT	PL	SK	SL	BG	RO	GA
				x																		

Table 1 - Document version

The language used in the document and drawings is subject to contract negotiations with the User.

In case of “Translated Instructions”, the Manufacturer also provides the “Original Instructions” document.

DOCUMENT REVISION

The following table shows the current revision of the document:

Revision	Description	Prepared	Verified	Approved	Date
1	First Issue	B. Bellagente	D. Malnati	U. Bisotti	03-04-2023

Table 2 - Document revision

GLOSSARY

The following table lists the terminology adopted in the manual:

Termine	Descrizione
The Manufacturer	Ginger Additive s.r.l.s.
The Machine	Ginger.one
The User	End user of the Machine

Table 3 - Glossary

SYMBOLS

The following table shows the used symbols:

Symbol	Description
	REQUIREMENT: Warning about the obligation to perform a given operation specified by further symbols and/or warnings.
	FORBIDDEN: Warning about the absolute prohibition on performing a given operation, specified by further symbols and/or warnings.
	DANGER: Warning about the danger arising from performing a given operation, specified by further symbols and/or warnings.

Table 4 – Symbols

PROFESSIONAL PROFILES

The following table lists the professional figures considered in using the machine as described in the present manual:

Symbol	Professional profile	Description
	Installer	Professional possessing specific technical knowhow who performs the machine installation and the electrical, hydraulic, and pneumatic connections
	Operator	Qualified figure utilising the machine for their own and permitted uses; authorised to clean the machine
	Maintenance Technician	Qualified figure able to carry out ordinary or extraordinary maintenance of electrical, hydraulic, or pneumatic nature following the instructions issued in the present manual. They are also authorised to clean and disassemble the machine.

Table 5 – Professional Profiles

	FORBIDDEN: the use of the machine is forbidden to any person, professional or non-professional, except for the ones explicitly listed.
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1. IDENTIFICATION

1.1. Plaque and type designation

The following figure shows the layout of the plaque on the machine:

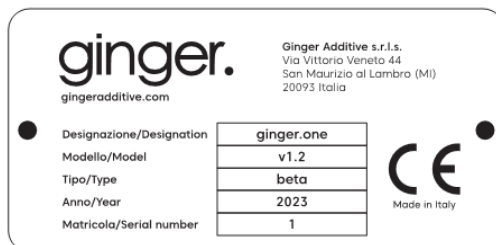


Figure 1 - Machine Plaque Layout

The following table shows the machine identification data included on the plaque:

Data	Description
Designazione/Designation	Official name of the machine
Modello/Model	Machine model
Tipo/Type	Machine type
Anno/Year	Machine build year
Matricola/Serial number	Machine serial number

Table 6 - machine identification data

	REQUIREMENT: Should the plaque detach from the machine, request a copy of the plaque from the Manufacturer, sending the informations contained in this manual.
--	---

1.2. Manufacturer name and address

The following table shows the data relating to the Manufacturer:

Data	Description
Name	Ginger Additive s.r.l.s.
Address	Via Vittorio Veneto 44 20093 San Maurizio al Lambro (MI) – Italia
Phone	+39 346 6266578
www	www.gingeradditive.com
e-Mail	info@gingeradditive.com

Table 7 - Manufacturer data

1.3. Warranty

Refer to the "Testing Contract - ginger.one v1.3 beta" agreed to at the purchase of the machine.

1.4. Service life

The expected service life of the machine is estimated as 10 years since first activation. Such estimation is to be considered valid exclusively if all requirements included in this manual are fulfilled.



REQUIREMENT: Comply with every warning issued in the present manual, both regarding the usage of the machine and its cleaning and maintenance.



FORBIDDEN: Do not allow usage of the machine to unauthorised personnel, personnel that did not receive the appropriate training , or personnel not possessing the skills necessary to the use and maintenance of the machine.



WARNING: Improper use of the machine can be dangerous to persons and/or objects; refer to the remaining risks described in this manual for specific dangers arising from an improper use of the machine

1.5. Declaration of Conformity CE

The machine referred to in this manual is subject to the application of the Machinery Directive 2006/42/CE, as such, the Declaration of Conformity CE is drafted (see Attachment 1) .

1.6. Applicable Standards

The following table shows which European Directives are applicable to the machine:

Directive	Description
2006/42/CE	Machinery Directive
2014/35/CE	Low Voltage Directive (its application is implied in the aforementioned Machinery Directive application, therefore it is not reported in the Declaration of Conformity CE)
2014/30/CE	Electro Magnetic Compatibility

Table 8 - applicable European Directives

The following table shows which norms are applicable to the machine:

Standard	Description
ISO 12100: 2010	Safety of Machinery - General Design Principles - Risk Assessment and Risk Reduction
ISO/TR 14121-2:2013	Safety of Machinery - Risk assessment - Part 2: Practical guidance and examples of methods
EN 60204-1: 2018	Safety of machinery - Electrical equipment of machines - Part 1: General requirements

Table 9 - Applicable Norms

1.7. Other Machinery and its Declaration of Conformity

There is no other machinery included inside the machine for which Declarations of Conformity CE have been issued in compliance with the Machinery Directive.

1.8. Partly Completed Machinery and their Declaration of Incorporation

There is no other machinery included inside the machine for which Declarations of Incorporation have been issued in compliance with the Machinery Directive.

2. GENERAL DESCRIPTION

The following image shows the machine:

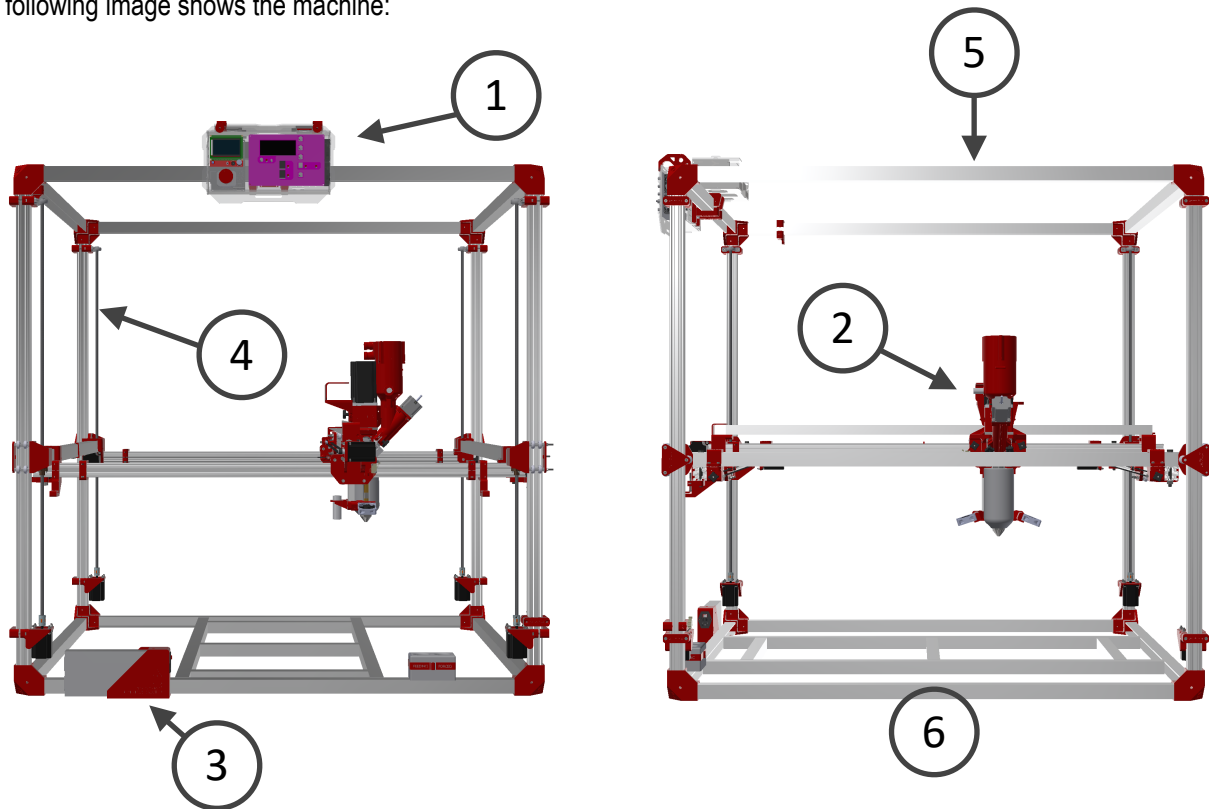


Figure 2 – Ginger.one

The following table lists the components of the machine:

Component	Description
1	Operation Control Panel
2	Extrusion head
3	Electrical Supply and controlled socket
4	Vertical movement axis
5	Machine frame
6	Loading/Unloading side

Table 10 - Components of the machine

Refer to Attachment 2 for the size diagram of the machine.

Refer to Attachment 3 for the electrical diagram of the machine.

Refer to Attachment 4 for manuals and data sheet of the individual components of the machine.

2.1. Control Panel

The following image shows the control panel:

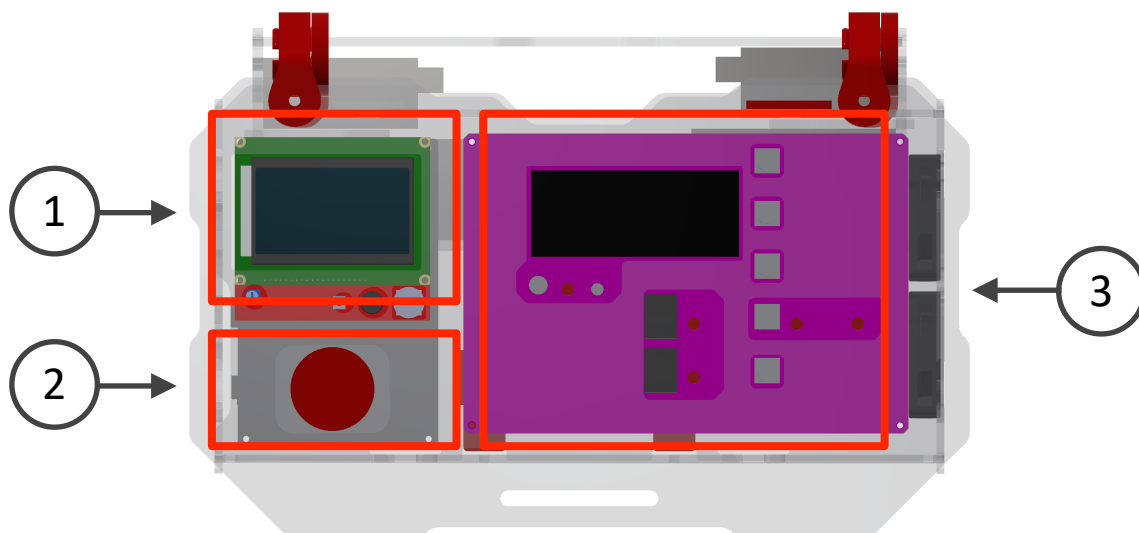


Figure 3 – Control Panel

The following table shows the components of the control panel:

Component	Description
1	Kinematics and Print Control Panel
2	Emergency button
3	Extruder and Print temperatures Control Panel

Table 11 - Components of the Control Panel

2.2. Extrusion Head

The following image shows the material extrusion head:

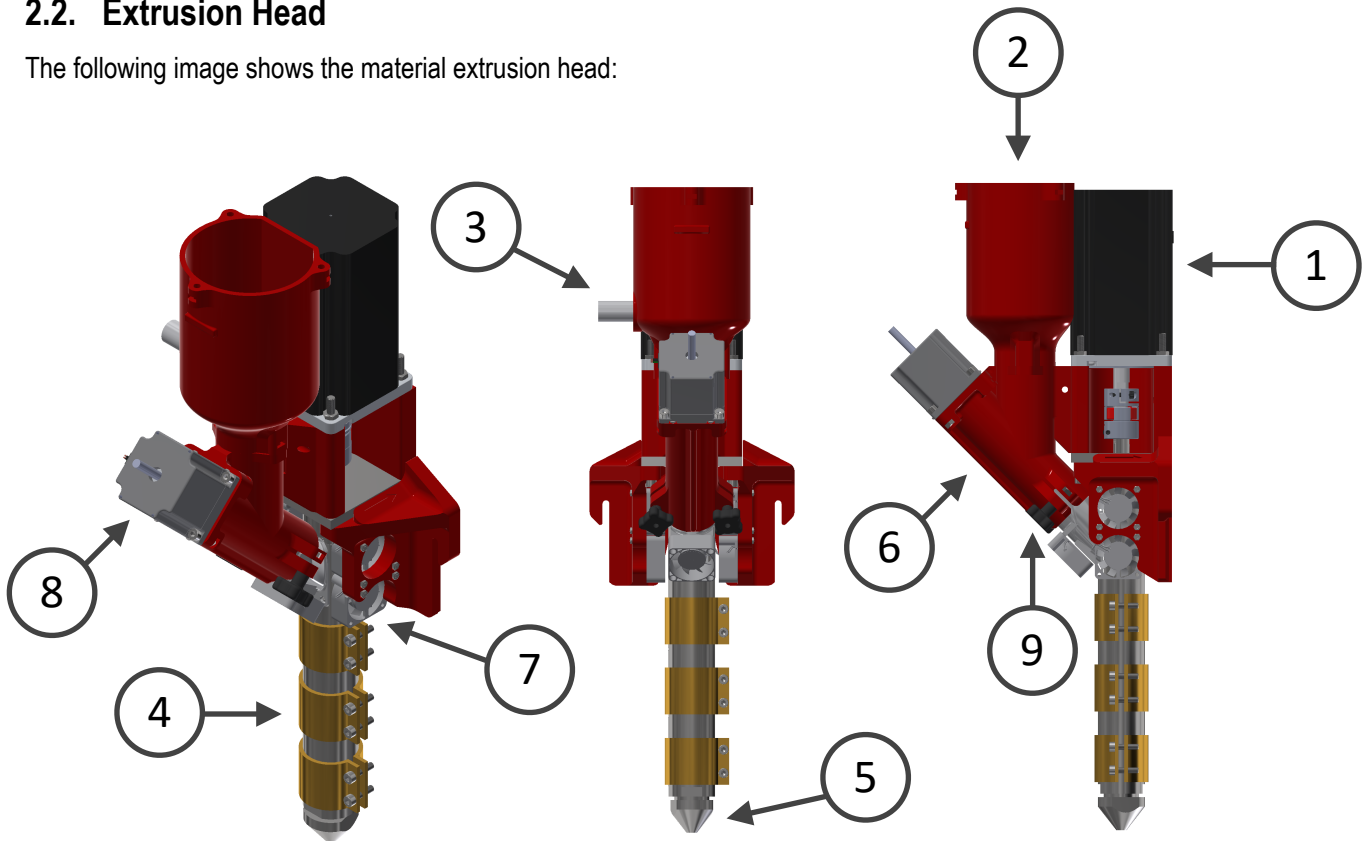


Figure 4 – Extrusion Head

The following table lists the components of the extrusion head.

Component	Description
1	Principle Motor
2	Hopper
3	Load Sensor
4	Heated Barrel
5	Nozzle
6	Feeding Auger
7	Cooling Fans
8	Auger Motor
9	Knobs for hopper removal

Table 12 - Components of the extrusion head

2.1. Intended Use

The machine is designed and built to 3D print objects of various shapes, using thermoplastic polymers exclusively, including:

- PLA
- PETG
- PP
- HDPE
- EVA
- Virgin Materials, , shredded or fibre reinforced

The extruder is designed to work with diverse thermoplastic materials. The good result of the printed piece however depends on the material and its degree of polymerisation, and by the ability of the operator in programming printing parameters.



FORBIDDEN: It is forbidden to use the machine for uses different from the specified one: incorrect use could result in damage to people and/or objects.

2.2. Usage Limits

Any usage of the machine that differs from the one described in paragraph 2.1 is to be considered incorrect and as such it is strictly prohibited.



FORBIDDEN: Usage of the machine is only allowed to qualified and authorised personnel; usage of the machine by any other person is forbidden.



FORBIDDEN: Any modification to the layout of the machine without the manufacturer authorisation is strictly forbidden.



FORBIDDEN: The machine is structured to be used by only one operator during the operation phase.



DANGER: It is dangerous to tamper with the machine and its accessories in order to modify the intended use



DANGER: Any usage of the machine that differs from the one described in paragraph 2.1 INTENDED USE can lead to damages to people and/or objects

2.3. Dimensions and Weight

Overall weight of the machine <100Kg.

For the dimensions of each component of the machine, refer to the attachments.

2.4. Environmental Conditions

The following table lists the Environmental Conditions for use of the system in FRT test:

Condition	Value
Ambient temperature for operation	(0÷+45) °C
Ambient temperature for storage and shipping	(-25÷+70) °C
Maximum temperature gradient	1,1 K/min
Humidity Class	EN 60204 point 4.4.3
Atmosphere	Libera da sostanze aeriformi infiammabili, esplosive o corrosive

Table 13 – Environmental Conditions for use of the system

It is required that the parts of the system are placed in a covered and dry place.

2.6. Connections

2.6.1. Electrical connections

The following image shows the connection point of the electrical network of the machine:

The electrical connection requires **220 VCA (50 Hz)**.

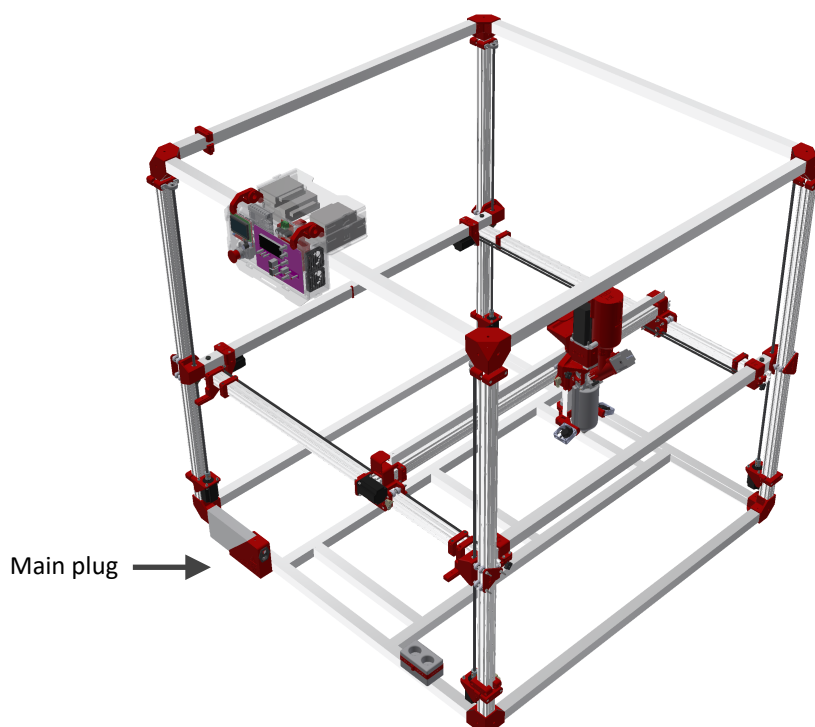


Figure 5 – Network coupling point

2.6.2. Connessione al PC

The following image shows the connection point of the machine with a PC:

The PC connection requires a **USB-b** cable.

From this port it is possible to control the kinematics of the machine and start a print file.

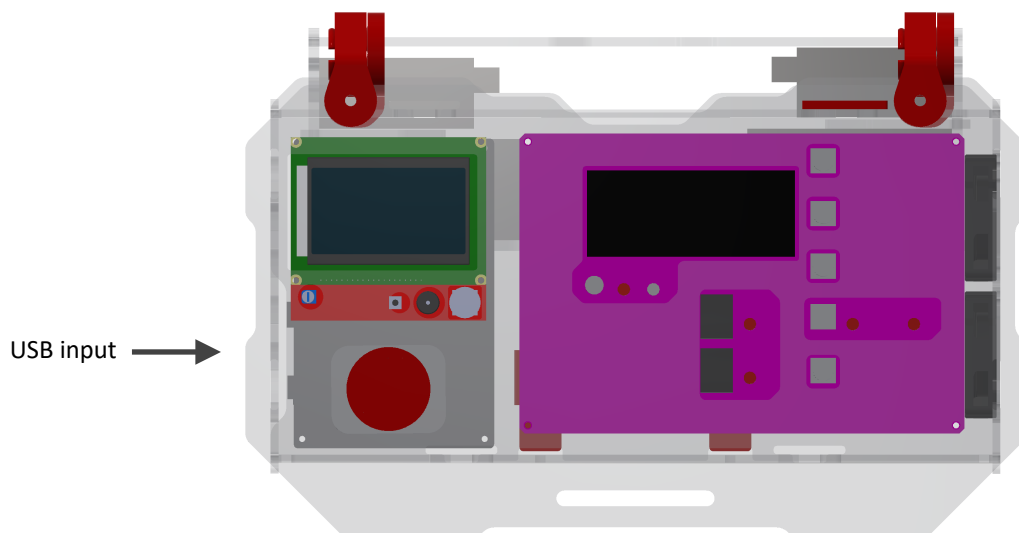


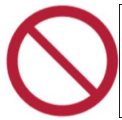
Figure 6 – Coupling point for PC

3. SAFETY

3.1. General warnings



Warning: The operator is required to carry out a training period for the usage of the machine



FORBIDDEN: Unauthorised and unqualified personnel must not put into operation, regulate, or repair the machine. Refer to this manual for any necessary information.



FORBIDDEN: Verify that there are not any unauthorised personnel in the vicinity before putting the machine into operation



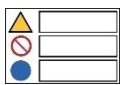
FORBIDDEN: Do not tamper with the electrical system, the pneumatic system, or any other system. Do not remove safety devices



FORBIDDEN: In case of pacemaker use, verify its compatibility with the machine



FORBIDDEN: Do not wear rings, wrist watches, jewellery, torn or dangling clothes such as ties, scarfs, unbuttoned jackets, or any garment that might get caught in parts of the machine



WARNING: Operate on the machine or in its vicinity taking into account what is reported on the labels attached to it



REQUIREMENT: The operator must have read the present manual with care and must have acquired the due knowledge on the technical specifications and commands of the machine, before proceeding in putting the machine into operation



REQUIREMENT: Be sure that all safety devices (barriers, protections, casing, microswitches, etc.) have not been tampered with and that they are perfectly functioning, prior to operation; otherwise, ensure their restoration



WARNING: The Manufacturer declines any responsibility for any damage to people and/or objects due to failure to comply either with requirements relating to safety or the rules contained herein



WARNING: In case of doubt, contact the technical support service of the Manufacturer

3.2. Emissions



WARNING: The acoustic pressure value measured in operation is less than 80dB(A)



WARNING: On the machine there are no sources of:

- Radioactive material or ionised radiation
- Laser radiation
- Non ionised radiation
- Biohazard material

3.3. IP Code

No specific IP grade is required: the machine is provided with individual parts equipped with their own IP grade.

3.4. Electromagnetic Compatibility

The Electromagnetic Compatibility Directive is applied to the Electrical system and to every electric/electronic parts underlying it.

3.5. PPE – Personal Protective Equipment



REQUIREMENT: The operator must wear suitable head protection



REQUIREMENT: The operator must wear suitable hand protection



REQUIREMENT: The operator must wear suitable foot protection



REQUIREMENT: The operator must wear suitable body protection



REQUIREMENT: The operator must wear suitable eye protection

3.6. Environments with risk of explosion



DANGER: The Machine is not suited to be used in environments with risk of explosion

3.7. Residual risks

The following table lists residual risks present on the machine:

Condition/Part	Risk factor	Developed solutions
Axis movement area	Danger of crushing. 	Adequate PPE use Read the manual Verify protections
Material extrusion zone	Danger of burning 	Adequate PPE use Read the manual Verify protections
Control panel zone and electrical coupling zone	Danger of electrocution. 	Adequate PPE use Read the manual Verify protections

Table 14 – Residual risks



FORBIDDEN: The machine is built to be used only by one operator during operation: the presence of two operators is strictly forbidden during operation.

3.8. Protections

3.8.1. *Safety design measures*

The machine has been designed applying the following safety design measures:

- Electrical system built in accordance with norm EN 60204-1
- Hot parts protected by the appropriate shielding
- Control panel position ergonomically suited for usage by the operators
- Mechanical parts sized in accordance with applicable norms

3.8.2. *Protection measures*

The machine is provided equipped with the following personnel safety devices:

- Emergency buttons chain

3.8.3. *Warning Labels*

The following image shows the locations of warning labels present on the machine:

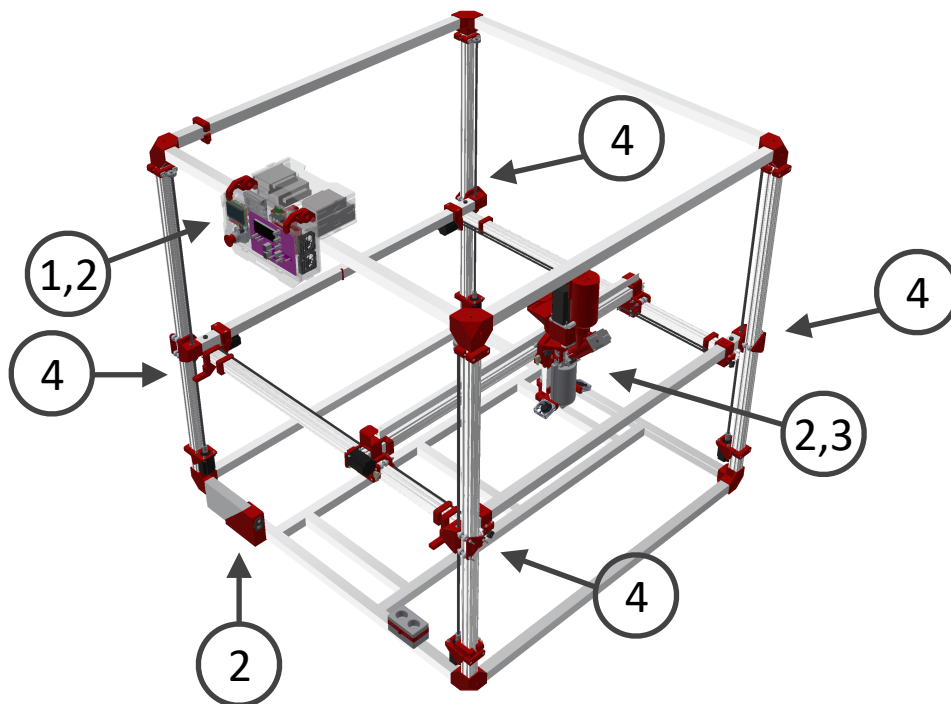


Figure 7 – Warning labels location

The following table lists the warning labels present on the machine:

N°	Symbol	Description
1		Mandatory use of PPE during installation, setup, maintenance, and cleaning phases
2		Electrical current danger
3		Hot surface dangerr
4		Hand crushing danger

Table 15 – Warning labels

	REQUIREMENT: In case of detachment and/or loss of the labels, replace them with commercial equivalents
---	---

3.9. Operating method in case of injury, failure, or fire

How to act in the event the machine is **STUCK**

- Remove the cause highlighted by the software
- Select the reset command from the software
- Verify that the machine works safely and resume normal activity

How to act in the event of **INJURY** during machine usage:

1. Stop the machine by pressing the emergency stop button
2. If the injured is trapped in parts of the machine they must not be moved and emergency health services and fire department must be called
3. if the injured is not trapped in parts of the machine, call emergency health services

How to act in the event of machine **FAILURE**:

1. Stop the machine by pressing the emergency stop button
2. Consult the maintenance personnel of the establishment and/or contact the Manufacturer Service

How to act in the event of machine **FIRE** and hydraulic oil fire:

1. in the event of a fire, remove voltage from the machine disabling the main switch of the electrical system, located behind the machine
2. Call the fire department and activate evacuation procedures laid down by the safety plans of the company, being mindful of any required PPE
3. In the event trained and specialised personnel in managing fire emergencies and first aid is present:
 - Perform any possible control and extinguishing of the outbreak safely and with the available means
 - Remove any injured persons



FORBIDDEN: Do not use water to extinguish fires on the machine while voltage supply is active



DANGER: Extinguishing a fire in progress is a complex operation that can only be carried out by specialised personnel (fire department), equipped with adequate gear

4. MACHINE PREPARATION FOR USE



REQUIREMENT: The personnel performing transport, handling, and installation operations must be equipped with garments and PPE, as prescribed by the norms in force in the operating country



FORBIDDEN: Use adequate devices (barriers, warning signals, etc.) to make sure the access to the lifting operations zone is forbidden to people not directly involved in the operation



WARNING: The lifting of the machine or its parts must be performed by specialised and authorised personnel (forklift operators, crane operators, etc); in the event that the size does not allow sufficient visibility, a person on the ground signalling any obstacles is necessary



REQUIREMENT: Both for safety and the correct functioning of the machine it is essential that the machine be connected to a grounding system that meets all the requirements of industrial rules for the prevention of injury and is compliant to what is required by the existing national regulatory frameworks in the country of operation of the machine



WARNING: The Manufacturer declines any responsibility for any damages to people and/or objects caused by the disregard of safety measures related to lifting and movement of materials, be it inside or outside of the establishment

4.1. Transport

Transportation of the machine is done using a wooden case that can be handled by pellet truck.

4.2. Storage

To store the machine, comply with the following rules:

1. Position all components in enclosed space, sheltered from the weather
2. Cover the machine components with appropriate plastic sheeting or otherwise protective material

4.3. Installation and assembly



WARNING: The installation of the machine **IN BETA VERSION** must be performed in the presence and with the supervision of the Manufacturer

5. OPERATING INSTRUCTIONS



WARNING: Only one operator is allowed during machine operation



REQUIREMENT: The operator must follow a period of training for usage of the machine



REQUIREMENT: The operator must have read the present manual with care and must have acquired the necessary knowledge of the technical specs and the commands of the machine, before proceeding with putting the machine into operation



FORBIDDEN: Unauthorised and unqualified personnel must not put machine in operation, regulate it, use it or repair it. Refer to this manual for any necessary information

5.1. Machine start-up

Use the main switch to supply power to the machine:

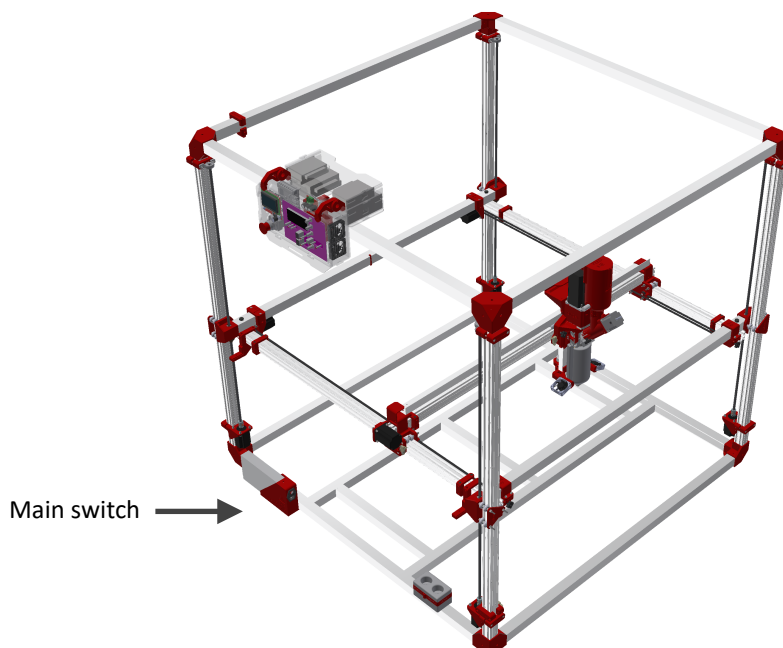


Figure 8 – Main switch

If the machine is properly connected to the power supply the two screens on the control panel will turn on and show their respective home screens:



Figure 9 – Control panel screens

5.2. Interface (HMI)

5.2.1. Extruder and print temperatures management

The following image shows the temperature management control panel:

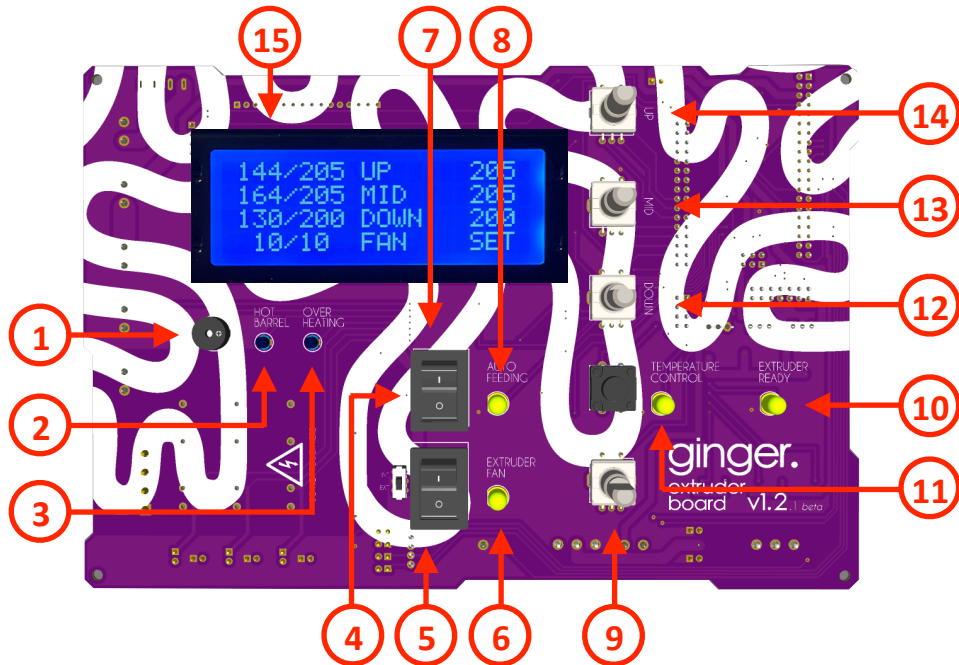


Figure 10 – Temperature management control panel

The following table lists the components of the control panel:

N°	Description	N°	Description
1	Sonic buzzer	9	Knob for EXTRUDER COOLING FAN speed setting
2	Red LED: HOT BARREL (at least one of the three zones UP, MID, and DOWN has surpassed 70 °C)	10	Green LED: EXTRUDER READY (nominal temperatures of the three zones)
3	Red LED: OVER HEATING (emergency caused by excessive temperature in at least one of the three zones UP, MID, and DOWN)	11	Blue LED: TEMPERATURE CONTROL (active temperatures control)
4	0/1 AUTO FEEDING switch	12	Knob for temperature setting in zone DOWN
5	0/1 EXTRUDER FAN switch, Selector INTERNAL / EXTERNAL	13	Knob for temperature setting in zone MID
6	Blue LED: EXTRUDER FAN (the cooling system is active)	14	Knob for temperature setting in zone UP
7	Blue LED: AUTO FEEDING (automatic pellet tank/extruder transport control activated) - it is activated only if (4) is positioned on "1" and "11" is turned on	15	Panel screen
8	TEMPERATURE CONTROL switch		

Table 16 - Components of the temperature control panel

The temperature management panel allows the control of the following temperatures:

- Three “UP” temperatures: temperature of the part underneath the pellet input
- One “MID” temperature: temperature in the central part of the extruder head
- One “DOWN” temperature: temperature of the extruder nozzle
- The temperature of the printing piece cooling fan

The three zones identified by the aforementioned temperature measures each have a thermistor and a heating element which allows for separate handling, to create the temperature gradient necessary for the correct extrusion of the material.

The following image shows the controlled and measured temperatures:



Figure 11 – Controlled and measured temperatures

The following table lists the elements in the panel:

Color	Description
Red	Instantaneous reading of the temperature in the three zones
Orange	Setpoint value in the three zones
Yellow	Value to be set by the lateral knobs (12), (13), and (14) Valore da impostare mediante manopole laterali (12), (13) e (14)
White	Instantaneous reading and setpoint of the percentage of PWM power relative to the cooling fan of the extrusion system (the setpoint is set with knob (9))

Table 17 - Temperature panel elements

5.2.2. Kinematics and print control panel

The firmware loaded on the board of this printer is Marlin 2.1.1 with custom configurations.

All of the relevant documentation is available at the following link:

<https://marlinfw.org/docs/basics/introduction.html>

The following image shows the main screen of the kinematics and print control panel:

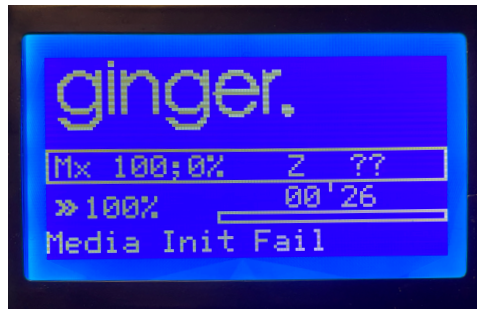


Figure 12 – Kinematics and print control panel

The main screen contains the following parameters (in order):

- Mixer value (speed ratio between auger and main extruder motor)
- Feedrate percentage (percentage of print feed speed)
- Current Z height
- Print time and cursor
- Various display communications

The following MENUs are also available (clicking on the knob):

- Motion (kinematics control)
- Mixer (mix value setting)
- Configuration (movement parameters configuration)
- Tune (with submenus Flow, Linear advance, Zeta offset)
- Retraction
- Bed levelling

5.3. Extrusion temperature setting

Set the operating temperature via knobs (12), (13), (14) and press the temperature control button. As soon as it is pressed the indicator on its right will light up BLUE (11), this signal corresponds to the start of the heating of the zones (up, mid, down).

It is always possible to modify SET, even once turned on. To change operating temperature, rotate the three lateral knobs, select the temperature and then click to set the new operating temperatures.

When at least one of the zones goes above 70°C the RED indicator (2) will light corresponding with the text “hot barrel”.

Once regime is reached the GREEN indicator (10) will light corresponding with the text “extruder ready”.

5.4. Material Change

When the machine is off, or with “automatic feeding” is off.

Unscrew the two knobs (9) on the extrusion head, without removing them, to loosen the attachment of the hopper being mindful of the connected cables. Place a basin under the hopper and pull the hopper out so that all pellets will flow into the basin. Reconnect the hopper.

Alternatively, vacuum the pellets from the top of the hopper.
Heat up the extruder e drain the material present in the hot barrel.
Add the new pellets and drain until the extrusion of the new material.

5.5. Changing the extrusion nozzle

Heat up only the zone “down” until the temperature is above the melting point of the polymer placed in the extruder.
Loose the nozzle with key and second key without removing it.

Wear thermal protection gloves and completely unscrew the nozzle. Place it somewhere where it will not be dangerous considering its momentarily high temperature.

It is recommended to clean the nozzle from the plastic while it is still hot.

5.6. Printing

5.6.1. *Creating the .gcode file*

1. Create a CAD model with the desired geometry.
2. Load the CAD model (STL model) on the slicing software and generate the G-code. In the software interface make sure that the geometry is contained within the print volume and that the printed object will not create interferences with the extruder.
3. Check start code and end code and save the .gcode file.

5.6.2. *Loading files from micro SD card*

Save files on the SD card and connect it in the dedicated slot on the left of the control panel.

5.6.3. *Loading files from USB cable*

Download management software for 3D printers control. There are several available: “Repetier host” and “Pronterface” are the suggested choices. Connect the computer to the printer control panel via the USB-b cable, in the USB slot on the left side, above the SD slot. Connect the printer to the software and send the printing program.

5.6.4. *Pre-printing routine*

4. Based on the estimate provided by the slicing software, verify that there is sufficient material available.
5. Turn on the printer using the power supply button.
6. Set the desired temperature values on the extruder control board display, turn on the temperature control pressing the temperature control button and verify that the blue LED has turned on.
7. Before selecting the file make sure that there are no objects around the printer that might interfere with the correct movement of the gantry. Distractions such as leaving the adhesive spray can on the side of the printer can cause damage during the auto home process such as the de-calibration of the printing plate and the breakage of the joints of the frame. It must also be verified that there are no traces of plastic drained from the nozzle that could prevent the correct reading of the probe, causing the same problems. Once this has been established go to **menu>print from media** and subsequently select the printing file.

8. Once the file has been selected the auto home process will start in which the extruder will place at the coordinates 0,0,0 corresponding to the left angle of the front panel. During the entire process it is necessary to control that there are no interferences that might stop the correct functioning of the endstops and of the probe. Once the auto home process is completed two beeps will sound and the screen will prompt to press the menu button. At this point it is possible to verify the z-offset specified in the main screen. If the measurement is not correct the z-offset can be set from **menu>motion>bed levelling>Z offset** and rerun the printing file selection procedure.
9. Once the auto home process is completed the display will show **“Check if extruder is ready”**, it will ask to confirm by pressing the knob, before doing so make sure that:
 - a. The reading temperatures of the extruder have reached their operating value
 - b. The nozzle diameter is the correct one.
 - c. If the auto feeding system is setup check if the power supply system tubes are placed correctly.
 - d. There are no interferences inside the printing area that block the extruder, furthermore it is necessary to check that there are no impediments in the movement of the gantry along the z axis.
10. Start the print.
11. During the brim printing, make sure that the material is not damp and that all the set parameters (printing speed, flow, temperature) are suitable for the given conditions.
12. Let the piece get printed.

5.7. Printing table calibration

5.7.1. First calibration of the frame

Check that the nozzle is clean and does not show plastic burrs. Check that there are no objects blocking the XYZ movement of the machine.

With machine off, move the vertical handling worm screws so that the gantry results levelled compared to the ground. (not necessary after the correct installation by technicians)

5.7.2. Automatic calibration via Capacitive probe sensor

Check that the sensor is mounted correctly according to the length of the nozzle in use, so that the sensor in the "deploy" mode is lower than the extrusion tip and that the sensor in the "stow" mode is higher than the extrusion tip. You can use the bl-touch submenu to deploy and stow. It is possible to use the "E19" shims printed in 3d for the correct assembly.

If correctly assembled, start an AUTO-HOMING and check the ZERO position of the zeta axis. After that it is possible to set a z-offset so that the nozzle is flush with the table in the zero position. **Menu>motion>bed leveling>Z offset**. Save the z-offset value in the EEPROM using the STORE SETTINGS command.

Start the automatic calibration from kinematics control inside the "bed levelling" menu or from start code. This will perform a sampling of the plane via a proximity sensor and create a mesh of the plane that will be used during printing.

The sensor sensibility can vary depending on the material and on external conditions such as temperature and humidity.

5.8. Adhesion to the plate

5.8.1. *Adhesion with flat plane and lacquer*

First method of adhesion to the printing plate (porcelain stoneware printing bed): Clean the MDF plane and make sure that there are no foreign bodies such as pellets or screws before positioning the bed, free the frontal side from the tie-rods and cautiously position the first slab of 60x120 stoneware. Once placed, insert the second slab making sure not to hit the previous one, position slowly the two slabs and make sure they cover the entire needed printing area, apply the adhesive spray on the printing bed and let it dry for a couple of minutes.

5.8.2. *Adhesion with MDF and mechanical fixing*

Second method of adhesion of the plate (MDF printing bed): in this case the print can be performed directly on the MDF bed.

Make sure to prepare the printing file (G-code) implementing a brim at the base of the object sufficient enough to be mechanically fixed to the printing plate. Following the print of the brim pause the printing process (menu>pause print) make sure that the printing is paused and that there are no moving parts, move the extruder (menu>motion>move axis) so that it does not interfere with the fixing process, remove the frontal tie-rods and cautiously fix the brim to the MDF printing bed in the desired spots using a screwdriver and self-tapping screws, once fixed make sure that the screws are not interfering with the nozzle during printing, mount the frontal tie-rods. Make sure the bed is free from objects that might interfere with the printing process and click resume print.

5.8.3. *Adhesion with printing plates made of other materials*

It is always possible to use printing plates made of polymers that have a strong enough adhesion with the printing material. In this case it is possible to clamp them or fix them mechanically to the MDF plane. Always make sure that the clamps and fixing are not clutter impeding the nominal movement of the printer.

5.8.4. *Adhesion with heated beds*

There is a special socket controlled in the PSU box called "BED" to use heated surfaces up to 1500W. It is possible to control the socket relay via a contact on the back of the box. CARRY OUT ALL RELATIVE ASSEMBLY MANEUVERS WITH THE MACHINE OFF.

5.9. Changes in parameters during printing

5.9.1. *Flow rate, feedrate and others*

Perform regular visual checks and if necessary, perform corrective actions modifying the printing speed, the flow rate, the linear advance (K factor), the operating temperature and the retraction.

5.9.2. *Babystepping*

It is possible to vary the reference of the end effector in-live using the "babystepping" function in the dedicated menu in the printer control panel. This allows to increment "in small steps" of 0.1mm the extruder position during printing, and thus allows to correct any errors or motor' skipping.

It is extremely useful during the first layer to calibrate the z height and correct a wrong calibration of the plane.

5.9.3. *Extruder fans*

Using the extruder control panel it is possible to pilot the cooling fans directed on the print. Use the knob (9), rotating and pressing it, to choose a fan speed value 0-100. it is possible to visualise the value on the screen. Press the switch (5) to activate the fan. Make sure that the LED (6) confirming activation is lit up.

The fans can be manually oriented through knobs on the extruder head. Do this if and only if the machine is off and the extruder is cold.

Usage of the fans is useful in the event the print necessitates cooling because the printing speed is such to not allow the natural cooling of the printing material. Or for specific features like elevated overhangs and small bridgings. It is not recommended to use the fans during the first layer, to guarantee a better adhesion to the plate.

It is possible to use the INT / EXT selector to the left of the switch (5) on the Extruder Board to drive the fans directly from the Motherboard. If positioned on "EXT" use the M106 command via gcode to modulate its power.

5.10. Loading of the material

5.10.1. Manual loading

During printing make sure that there is material loaded in the hopper. If the automatic pellet transport system is off it is always possible to activate the switch (4) to start the level sensor. This way, if the hopper gets empty, the system will beep to warn that the material has run out.

To load material: pause the machine from the control panel, move the axes so that the extruder head is close to the loading/unloading side (7), and manually load the hopper. Start the printing again from the control panel.

5.10.2. Automatic feeding

Before the activation of the machine, fix the pellet vacuum system on the extruder head.

Plug the vacuum to the power supply (6). Link the vacuum tube to the vacuum, link the feeding tube to the pellet basin. Always keep the vacuum ON. From the control panel activate the extruder switch (4) for auto-feeding control.

This way the system controls few seconds cycles during which the material gets fed from the basin to the hopper through the suction provided by the vacuum. In case of failure or too long cycles a beep will sound between a cycle and the next.

Do not use vacuums with power above >2kW.

5.11. Removal of the print

Once the print is done the temperature control automatically switches off, indicated via the blue temperature control LED shutting down, the end-of-print process continues by performing a z raise and auto-homing of the x and y axes, after this is 3 beeps will sound.

To remove the printed object wait for it and the extruder to cool off (hot barrel LED off), move the extruder to avoid hitting it unintentionally, remove the fixing screws in case of MDF printing plate or use a common carpentry spatula to remove the print from the printing plate this may take a few minutes and may require various action points where to use the spatula, it is important not to direct the spatula toward oneself, performing this operation while not paying attention may result in injuries.

It is now possible to move the printer and send printing programs from the computer.

5.12. Machine shutdown

To shut down the machine:

- Make sure the printing cycle is terminated
- **Wait for the extruder head to cool off !**
 - press the “temperature control” button so that the LED (11) turns off
 - wait for the red LED to turn off (which corresponds to reaching 70°C in every zone of the extruder)
- Shutdown the main switch of the machine

5.8. Emergency stop

In case of emergency, press the emergency stop button on the control panel. Pressing it immediately shuts down the machine movements, the extrusion, and the heating zones.

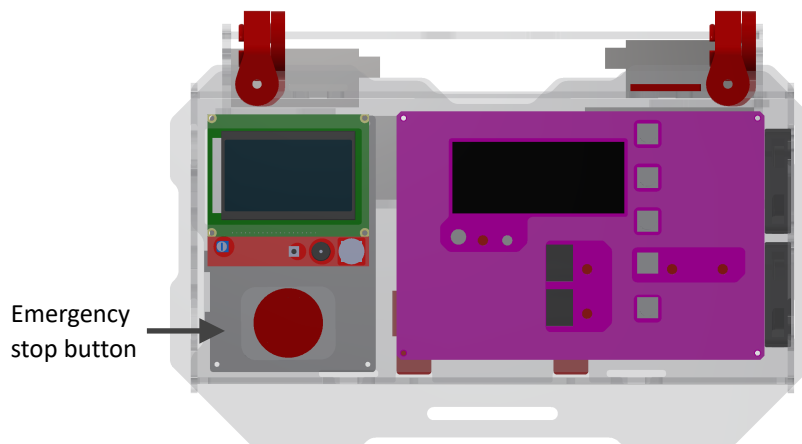


Figure 13 – Emergency stop button on the control panel



DANGER: Pressing the emergency stop button and shutting down the extrusion process does not entail the immediate cooling of the extruder. let it cool off before touching it and always use safety gloves to do so.

5.9. Updating the Firmware

5.9.1. Temperature control panel

Download the firmware file with .ino extension made by the manufacturing company. Download the software Arduino IDE and open the firmware file. Unplug electrical current from the machine!

Connect the computer to the USB port on the extruder board. “Build” the firmware and load it onto the board.

Unplug the USB cable (!) and turn on the machine.

5.9.2. *Kinematics and printing control panel*

Download the firmware file "firmware.bin" made by the manufacturing company. Extract the micro SD card from the board and copy the file onto it. Re-insert the micro SD card into the board and perform a reset.

It is not necessary to unplug the current from the machine.

5.10. Signals to pay attention to

The machine is equipped with an acoustic alarm that will sound in the following scenarios.

1. When there is an emergency overheating signal (quick pulsating sound). If and only if the temperature control is active
2. When the material in the hopper has run out or if there is a failure in the pellet transport (slow pulsating sound). If and only if the auto-feeding control is active.

6. MAINTENANCE AND CLEANING



WARNING: Contact the staff in charge of maintenance, without proceeding with further repairs, for any malfunctioning of the machine or damaged components



REQUIREMENT: Before starting maintenance operations it is necessary for the operator to wear adequate protective devices, as requested by the norms in the operating country



REQUIREMENT: The operations must be carried out swiftly at the first sign of anomaly, to avoid worsening of the problem and damage to further components



REQUIREMENT: Before starting the cleaning and/or maintenance of the machine and before removing any protection, make sure that the main switch is in the "OFF" position, so as to remove electrical supply from the machine during the performing of the operations



REQUIREMENT: Disable voltage on the line connected to the main switch, if there need to be operations performed on it or in its vicinities



REQUIREMENT: Use only electrical equipment that is correctly connected to ground and is compliant with the national safety measures



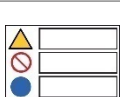
REQUIREMENT: The operations must be carried out swiftly at the first sign of anomaly, to avoid worsening of the problem and damage to further components



WARNING: In the event of irregularities or the malfunctioning of any component of the machine, check that during its use the directions outlined in the following paragraphs have been followed



FORBIDDEN: Do not wear rings, wrist watches, jewellery, torn or dangling clothes such as ties, scarfs, unbuttoned jackets, or any garment that might get caught in parts of the machine



WARNING: Make sure all labels are kept legible, so as to guarantee their complete visibility to the operator. Substitute any and all labels that for whatever reason may have deteriorated or that are not clearly visibile (refer to Chapter 3.8.3 WARNING LABELS)



DANGER: Do not perform rushed or makeshift repairs that might undermine the proper functioning of the machine

6.1. Ordinary Maintenance

The following procedure lists the checks to be ordinarily carried out:

- Cleaning of the manual loading hopper
- Cleaning of the material refuelling auger
- Check that all the controlled fans (board and extruder) are always on
- Check the tension of the horizontal transmission chains
- Check the wear of the wheels of the trolleys of the movement axes
- Check the tension on the tie-rods of the structure
- Greasing of the worm screws for the vertical movement
- Checking and cleaning the filter of the auto-feeding system

6.2. Troubleshooting, failure diagnosis and repairs

All the components of the printer that are 3D printed can, if necessary, be printed by the user and substitute after consulting the manufacturer. All the files relating to the parts are open-source and are part of the attached documentation.

For any possible failure please contact the manufacturer.

7. REPLACEMENT PARTS AND CONSUMABLE MATERIALS

The following table lists the replacement parts of the machine:

N°	Model	Code	Supplier
1	Wheels for v-slot trolleys	X-treme (polycarbonate)	Ginger Additive s.r.l.s.
2	Wheels for v-slot trolleys	Metal	Ginger Additive s.r.l.s.

Table 18 – Replacement parts

The following table lists the consumable materials of the machine:

N°	Model	Supplier
1	virgin PLA	Ginger Additive s.r.l.s.
2	recycled PLA	Ginger Additive s.r.l.s.

Table 19 – Consumable materials

8. DISMANTLING

There are specific legal provisions forbidding the abandonment or dispersion in the environment of components or materials that may directly or indirectly cause fires or pollution.

8.1. Destruction

At the end of the operating life of the Partly Completed Machine it is appropriate to proceed as the following:

- Remove the identifying label and conserve it along with the present manual and the manufacturer-issued documentation
- Remove the lubrication oil from the components or any parts that may contain it, separating it in an appropriate container
- Remove and separate the electrical parts (cables, buttons, connectors, programmer, boards, electrical motors, electrical and electronic components, etc.)
- Remove and separate the plastic parts (handles, screens, buttons, etc.)
- Separate the metal parts by type (steel, aluminium, cast-iron, bronze, etc.)

The aforementioned operations can be performed without the need of special equipment.

No special abilities are required of the operators.

Use safety gloves in removing and dismantling the metal parts.

8.2. Recycling

Not applicable.

8.3. Disposal

Contact specialised firms for the disposal of the different parts outlined in the previous paragraph.

Contact authorised firms for the disposal of the substances and chemicals employed in the lubrication of the machine.

9. ATTACHMENTS

The following table shows the attachments to this manual, of which they are an integral part; therefore the proper storage of all the documentation is recommended:

N°	Attachment name	Description
1	Declaration of Conformity - N.00	Declaration of Conformity CE
2	DRAWING	Size diagram of the machine
3	ELECTRIC SCHEMATIC	Electric diagram
4	COMPONENTS	Data sheet and components certificates

Table 20 - Attachments to the manual

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